

Folium

End-to-End Design and Implementation of a Location-Scoped Citizen Science App for Botanical Cataloging

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Abstract

Biodiversity monitoring at fine geographic scales faces a persistent gap between the availability of generic citizen science platforms and the specific needs of bounded locations — urban parks, nature reserves, university campuses, and municipal green corridors. This project presents *Folium*, a production-grade native mobile application designed to enable community-driven botanical cataloging at any defined geographic area. The system implements a multi-tiered consensus validation framework adapted from iNaturalist's Research Grade model, incorporating expert-weighted voting that maps onto community role hierarchies (Student, Expert, Professor) to yield scientifically credible observation status. Built on a full-stack architecture comprising Expo (React Native + TypeScript), Supabase (PostgreSQL with PostGIS extension), and an automated deployment pipeline via Expo Application Services (EAS) and Fastlane, the application was submitted to the Google Play Store in May 2026, with a pilot deployment at the UAB Campus (Cerdanyola del Vallès) scheduled for June 2026. The project demonstrates that a single developer can architect, build, and deploy a production-ready citizen science platform that satisfies both technical rigor and the legal obligations of mobile distribution (GDPR, Creative Commons BY-NC 4.0 licensing), at a total project investment of USD 25.

Keywords

Citizen Science · Botanical Cataloging · React Native · Supabase · GDPR · Mobile Deployment · Biodiversity Monitoring

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1. Description of the Problem

1.1. Biodiversity Monitoring in the Digital Age

The documentation of plant biodiversity at fine spatial scales — building blocks, pathways, green corridors, restoration plots — represents an unsolved problem in institutional ecology. Systematic botanical surveys are resource-intensive, expert-dependent, and inherently static: a survey captures a point in time, not a living record. The emergence of citizen science as a methodological paradigm has begun to address this limitation: distributed, community-sourced observation networks can generate continuous temporal coverage over extended geographic areas at a fraction of the cost of expert-only surveys. Platforms such as iNaturalist (California Academy of Sciences) have demonstrated this model at global scale, accumulating hundreds of millions of observations across taxa and geographies ¹.

However, the translation of this global model to bounded institutional contexts — a university campus, a nature reserve, a municipal park — exposes structural limitations in existing tools. Generic platforms are optimized for geographic breadth, not for the depth of engagement, role specificity, and institutional cohesion that a university-scale living laboratory requires.

1.2. Bounded Locations as Underserved Botanical Contexts

Fine-scale botanical documentation is most valuable — and most absent — at bounded locations where human activity and natural vegetation coexist: urban parks, periurban green belts, protected nature reserves, campus grounds, and municipal ecological corridors. These sites often support a mix of native species, ornamental introductions, and invasive taxa that together make them ecologically significant yet structurally heterogeneous. Despite this interest, species records at such locations are typically fragmented across departmental surveys, grey literature, and informal observation, with no unified, georeferenced, publicly accessible dynamic database.

This represents a missed opportunity both scientifically — for phenological research, invasion monitoring, and biodiversity benchmarking — and educationally, given the potential of these sites as accessible teaching resources for botanical field identification. The UAB Campus (Cerdanyola del Vallès, approximately 260 hectares of mixed Mediterranean scrubland, riparian corridors, and managed ornamental planting) was selected as the pilot deployment site for *Folium* precisely because it exemplifies this pattern: ecologically rich, spatially bounded, community-accessible, and lacking a structured live catalog.

1.3. Limitations of Existing Citizen Science and Botanical Platforms

An analysis of existing digital tools reveals that none satisfies the combination of requirements for a location-specific, community-driven, scientifically credible botanical catalog.

iNaturalist is the most advanced open citizen science platform available, offering consensus identification, geospatial visualization, and expert engagement. However, it is designed as a globally distributed system with no location-scoping capability. It cannot be configured to restrict observation scope to a defined geographic polygon, it does not support role hierarchies tied to community credentials, its gamification model is not

calibrated for place-specific cohort engagement, and its species database, while comprehensive, lacks location-specific contextual layers (site zones, managed areas, restoration plots) ¹.

- **Flora Catalana** (Societat Catalana de Biologia) provides an authoritative database of Catalan vascular flora but is a read-only taxonomic reference, not a contribution platform ².
- **Patrimoni Natural** (Generalitat de Catalunya) and the **Visor del Patrimoni del Solsonès** offer geospatial visualization of protected elements but have no citizen contribution model.
- The **Projecte Orenetes** (ICO) demonstrates an effective community observation model for birds, but its methodological design is taxon-specific and not extensible to botany ³.

Progressive Web Apps (PWAs), considered as a lower-cost alternative to native development, are technically incapable of delivering the GPS precision and camera hardware integration required for field-grade botanical observation. Web-based geolocation APIs operate with browser-mediated accuracy degradation that is unacceptable in forested environments where sub-plot precision is needed for valid spatial data. Similarly, browser-mediated camera access does not permit the sensor-level control necessary for consistent, high-quality specimen photography.

1.4. The Project's Origins and the Justification for Pivoting

The immediate predecessor to *Folium*, developed during the 2024–2025 academic year, consisted of a WordPress plugin implementing geospatial visualization of a campus botanical catalog (Leaflet.js + static `plantes.json`), complemented by a no-code local management application ⁴. This prior system was validated through a System Usability Scale (SUS) study with 90 participants, achieving a score above 94/100 — a result that validated the conceptual value of the catalog but also exposed the architectural ceiling of a single-administrator, web-centric model.

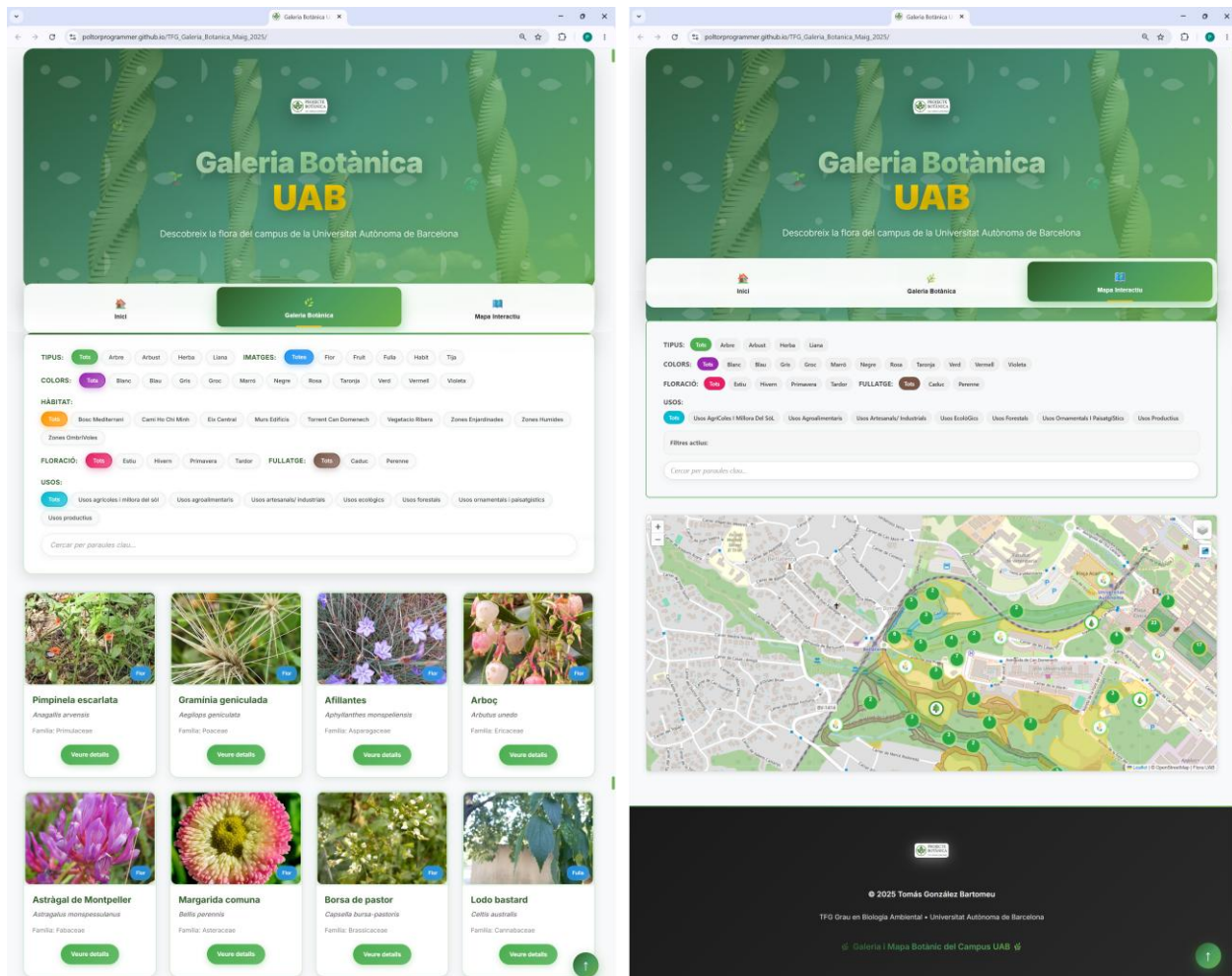


Figure 1: Interface of the predecessor system — Leaflet.js geospatial visualization of the UAB Campus botanical catalog, implemented as a WordPress plugin (2024–2025).

An initially planned deployment agreement with the Sant Cugat del Vallès municipal administration proved unachievable within the academic timeline. Rather than constraining the project to a static institutional dependency, this obstacle motivated a strategic pivot: from a “product for a client” model to a self-sustaining, location-agnostic citizen science platform, using the UAB Campus as the pilot Living Lab. This pivot expanded the project’s scope from data visualization to data generation, from administration to community governance, and from web deployment to native mobile distribution — each expansion introducing both technical depth and scientific validity.

1.5. The Identified Gap

The synthesis of the above analysis defines the specific engineering problem addressed by this project: the construction of a location-scoped, natively mobile, legally compliant citizen science platform that:

- Enables community observation contribution without expert-administrator bottlenecks.
- Implements a scientifically credible, expert-weighted validation model.
- Operates within a GDPR-compliant data governance framework.

- Is distributable via standard app stores at zero marginal operational cost. No existing tool satisfies all four conditions simultaneously.
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2. Objectives

The project pursues four objectives, ordered by increasing operational specificity:

1. **Develop a native mobile application** enabling any user to photograph, geolocate, and submit field observations of plant species within a defined area, incorporating AI-assisted species pre-identification via the PlantNet botanical identification API (Pl@ntNet, INRAE/CIRAD/INRIA/IRD) ⁵.
 2. **Implement a consensus-based observation validation system** adapted from the iNaturalist Research Grade model ¹, incorporating an expert-weighted role hierarchy (Student / Expert / Professor) to produce machine-readable observation status (pending / community_verified / scientifically_validated) suited for downstream scientific use.
 3. **Design and enforce a production-grade legal compliance framework** satisfying the requirements for distribution on the Google Play Store and Apple App Store, including: GDPR Article 17-compliant self-serve account deletion ⁶, a publicly hosted Privacy Policy, pre-permission GPS transparency onboarding, Creative Commons BY-NC 4.0 licensing for all user-generated content ⁷, and a community content moderation mechanism.
 4. **Submit the application to the Google Play Store and initiate a pilot deployment** at the UAB Campus (Cerdanyola del Vallès), establishing the technical and sociological viability of the location-scoped citizen science model and generating the observational dataset required for long-term botanical monitoring of the site.
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3. Material and Methods

3.1. Development Methodology

The project was executed following an **Agile iterative methodology**, organized in five sequential phases spanning January to May 2026 (Table 1). Each phase concluded with a demonstrable deliverable, reviewed against project objectives and against tutor feedback from two formally documented progress meetings (January 2026, March 2026) and one informal consultation (May 2026). Architectural decisions made during the early phases were documented as technical specifications prior to implementation, reducing costly mid-development pivots.

The entire codebase was developed by a single developer and maintained via Git version control (GitHub: PoltorProgrammer). No pre-existing mobile codebase was adapted; the Expo project was initialized from scratch in February 2026, ensuring architectural consistency and eliminating technical debt from the prior WordPress-centric prototype.

Table 1: Development phases, periods, and primary deliverables.

Phase	Period	Primary Deliverable
1. Pivot & Definition	January 2026	Citizen science model specification; tech stack decision
2. Core Infrastructure	February 2026	Database schema; Supabase configuration; Expo scaffold; authentication
3. Feature Development	March 2026	Geospatial map; species gallery; observation capture flow
4. Social & Gamification	April 2026	Consensus validation engine; social feed; user ranking and badges
5. Legal & Deployment	May 2026	GDPR tools; CC licensing; Play Store submission via Fastlane

3.2. Frontend Architecture: Expo (React Native) with TypeScript

Framework selection. The decision to use a native mobile framework rather than a Progressive Web App (PWA) was driven by three constraints specific to field botanical documentation. First, GPS accuracy in mixed woodland and built environments requires direct access to the device's native location subsystem; web-based geolocation, mediated by browser permission layers, introduces precision loss that is unacceptable for species-level spatial attribution. Second, high-resolution specimen photography requires direct camera sensor access and shutter control unavailable in a browser sandbox. Third, native rendering delivers the fluid, touch-optimized interface necessary for practical single-handed operation during field sessions.

Expo (SDK 51+) was selected over bare React Native for its managed workflow, which abstracts native module compilation and provides first-class hardware APIs ⁸. **expo-camera** enables high-resolution photographic acquisition with direct sensor access. **expo-location** provides sub-meter GPS coordinate extraction via the device's native location service. Both modules are essential for the scientific validity of the submitted observation data.

TypeScript 5.x in strict mode was applied throughout the codebase to enforce type safety across all taxonomic data structures, coordinate schemas, API response payloads, and database interface types. For a system in which data type errors could corrupt species records or observation coordinates, compile-time type enforcement is not optional.

Navigation is implemented via **Expo Router** (file-based routing), structuring the application around five primary tabs: *Interactive Map*, *Species Gallery*, *Observation Capture*, *Social Feed*, and *User Profile*. **NativeWind** (Tailwind CSS for React Native) provides consistent, responsive styling with a defined design system — botanical green (**#2D6A4F**), sage (**#95D5B2**), off-white (**#F8F9FA**), charcoal (**#2B2D42**), and amber (**#E9C46A**). **react-native-maps** renders hardware-accelerated Google Maps tiles with custom marker layers ⁹. **Supercluster** manages observation clustering at high densities, preventing map readability degradation as the dataset grows.

3.3. Backend Infrastructure: Supabase (PostgreSQL + PostGIS)

Technology selection. The prior static JSON data model (`plantas.json`) was incompatible with concurrent multi-user access, relational integrity between entities, and the dynamic validation state tracking required by the consensus engine. A relational database was mandatory. **Supabase** was selected as the platform for three additional reasons:

1. Its free tier provides the full range of services required — PostgreSQL database, authentication, object storage, and serverless functions — at zero cost, matching the project's budget constraints.
2. Its **PostGIS** extension enables native geospatial queries (observations within a polygon, proximity search, coordinate blurring for protected species) without additional middleware.
3. Its **Row Level Security (RLS)** framework enforces fine-grained access control at the database layer, eliminating entire categories of authorization vulnerabilities from application code [10](#).

Database schema. The relational model is organized around four primary tables:

- **profiles:** User identity (UUID primary key, 1:1 with Supabase Auth), public username, full name for scientific attribution, avatar image URL, academic `rank_level` (1: Student, 2: Expert, 3: Professor), cumulative `reputation_score`, system `role`, and total `observation_count`.
- **species:** The master botanical dictionary. Each record contains a GBIF global identifier for external cross-referencing, the scientific name (Latin binomial, unique constraint), localized common names in Catalan/Spanish/English, taxonomic family, binary flags for protected and alien status, a morphological `description`, and an `aggregated_traits` JSONB field that consolidates the consensus morphological characterization from all validated observations of that species. Protected species (`is_protected = true`) trigger coordinate blurring to a 5 km grid on retrieval, protecting sensitive populations while maintaining research utility.
- **observations:** The community data stream. Each record stores the submitting user reference, the identified species reference, a PostGIS `geography` coordinate, the validation `status` enum (`pending` / `community_verified` / `scientifically_validated`), a rank-weighted `confidence_score`, a `moderation_score`, JSONB `observed_traits` specific to the specimen, a JSONB `images` array (each element carrying URL, organ type, and vote counts), and a `parent_id` foreign key enabling conflict-fork trees when users disagree on identification.
- **validations:** Vote records with a composite unique constraint on (`observation_id`, `user_id`) enforcing the single-vote-per-user rule.

Two auxiliary tables (`media_votes`, `reports`) support photographic quality ranking and content moderation respectively.

Row Level Security policies are applied comprehensively: users can only write to their own profile and observation records, and read access to protected species coordinates

is filtered server-side. JWT tokens issued by Supabase Auth are validated on every database operation.

Storage buckets. Photographic assets are managed in two purpose-separated Supabase Storage buckets. `observation-images` is public-read and write-protected, using the naming convention `[scientific-name]_[organ]_[unique-id].jpg` (e.g., `rosmarinus-officinalis_flower_a1b2c3d4.jpg`). `user-profiles` stores avatar images, with a procedural fallback generating a 300×300 px pastel canvas with a centered nature emoji for users without uploaded photos.

3.4. AI-Assisted Species Identification

Species pre-identification is facilitated via integration with the **PlantNet API** (Pl@ntNet consortium: INRAE, CIRAD, INRIA, IRD) ⁵, which accepts photographic input and returns a confidence-ranked list of candidate species with their GBIF identifiers. This pre-identification is intentionally advisory: the system pre-populates the observation form with the top-ranked candidate, but the user reviews and confirms (or overrides) the assignment before submission. Final species attribution remains subject to the community validation process (Section 4.3), ensuring that systematic model errors do not propagate unchecked into the botanical database. This two-tier model — machine suggestion plus community consensus — mirrors the hybrid identification approach adopted by iNaturalist ¹ and is considered methodologically sound for citizen science datasets.

3.5. Build and Deployment Pipeline: EAS and Fastlane

Expo Application Services (EAS) Build manages the compilation of the TypeScript/JavaScript application bundle into platform-native binaries ⁸. For Android, EAS produces an **Android App Bundle (.aab)** processed via Gradle, incorporating all native module dependencies and signed with a production release keystore. For iOS, EAS produces an **IPA archive** via Xcode toolchain. The **Metro Bundler** optimizes JavaScript asset bundling and code splitting prior to native compilation.

Fastlane automates the submission pipeline: binary upload to Google Play Console, metadata synchronization (store listing text, screenshots, categorization, release notes), and release track management (internal → closed testing → production) ¹¹. This automation eliminates manual submission steps that carry a high risk of configuration error, ensuring reproducible deployments throughout the application lifecycle.

Over-the-Air (OTA) updates via EAS Update permit hot-patching of JavaScript assets without triggering a new store review cycle, enabling rapid post-launch corrections during the campus pilot phase without interrupting user availability.

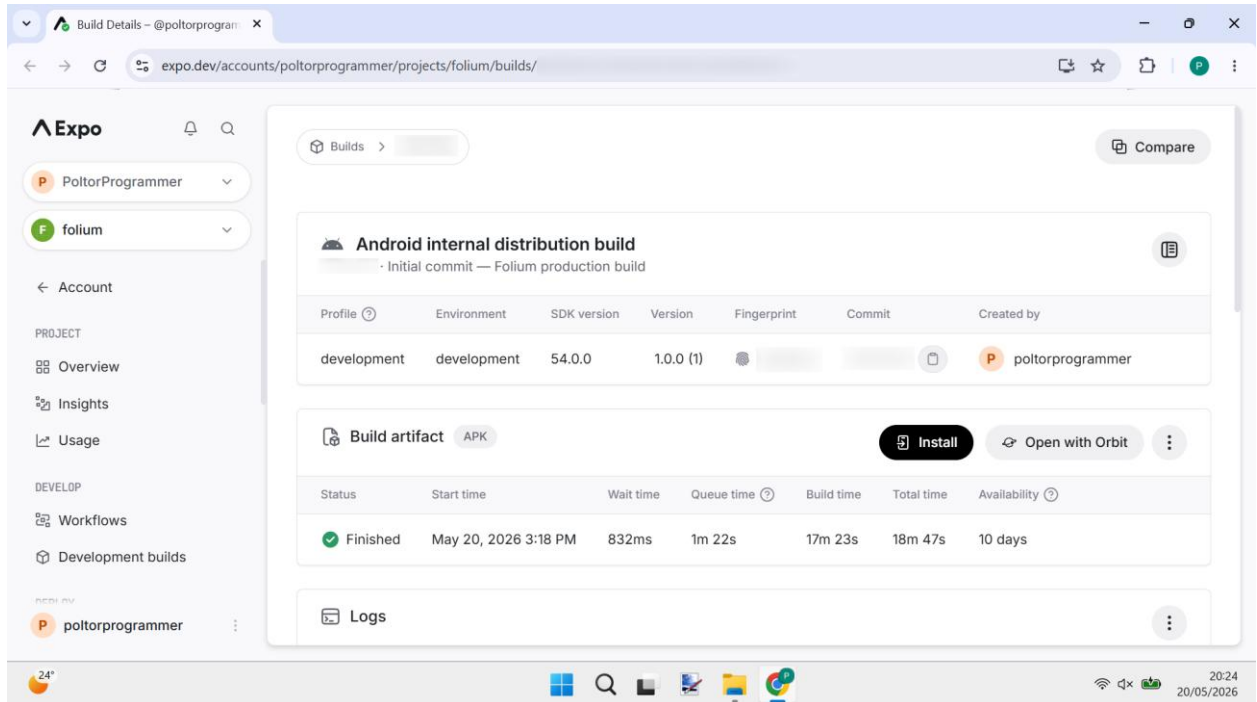


Figure 2: Expo Application Services (EAS) build console, showing build metrics, compilation configuration (SDK 54.0.0, development profile), and artifact generation details for the Android internal distribution build.

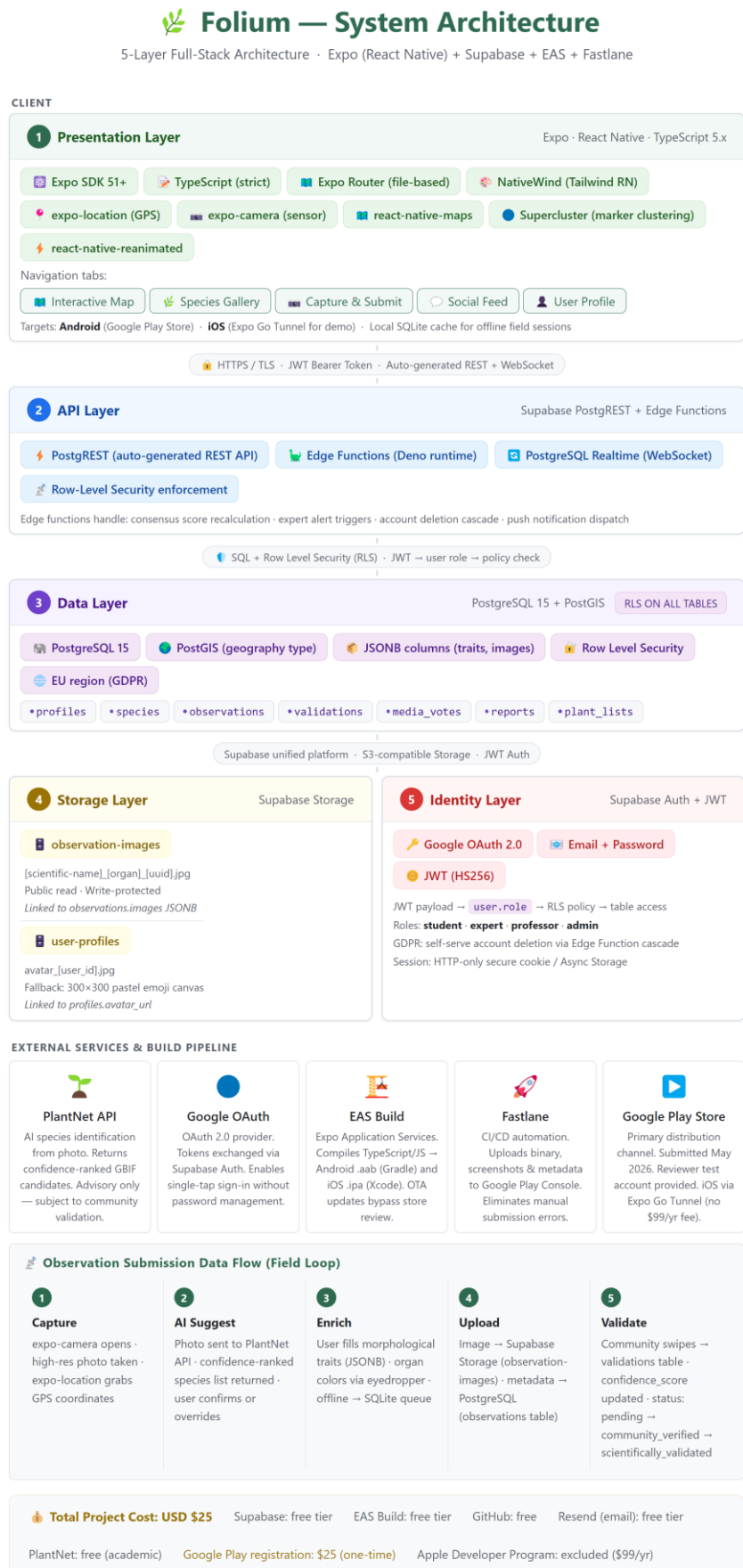
4. Description of the Solution

4.1. System Architecture Overview

Folium is organized as a five-layer architecture (Figure 3):

1. **Presentation layer:** Expo/React Native application, cross-platform (Android and iOS)
2. **API layer:** Supabase PostgREST auto-generated REST API and Edge Functions (Deno runtime) for complex serverless logic (expert alert triggering, consensus state machine)
3. **Data layer:** PostgreSQL 15 with PostGIS (Supabase-hosted, EU region)
4. **Storage layer:** Supabase Storage S3-compatible buckets for binary assets
5. **Identity layer:** Supabase Auth with JWT issuance; Google OAuth 2.0 and email/password providers

All inter-layer communication is encrypted via HTTPS/TLS. Production Supabase credentials are isolated in encrypted environment variables, separate from the development configuration, and are injected at build time by the EAS Build system.



4.2. Core Feature Modules

Interactive Geospatial Map. The application's primary exploration interface renders all validated observations as geolocated markers on a Google Maps base layer. Observations are clustered using Supercluster when spatial density exceeds visual clarity thresholds. Tapping a cluster zooms to reveal individual markers; tapping a marker opens an observation detail panel. The map supports filtering by species family, observation status, and date range. Protected species coordinates are blurred to a 5 km grid at the API layer, visible to all users but not revealing the precise location of sensitive populations.

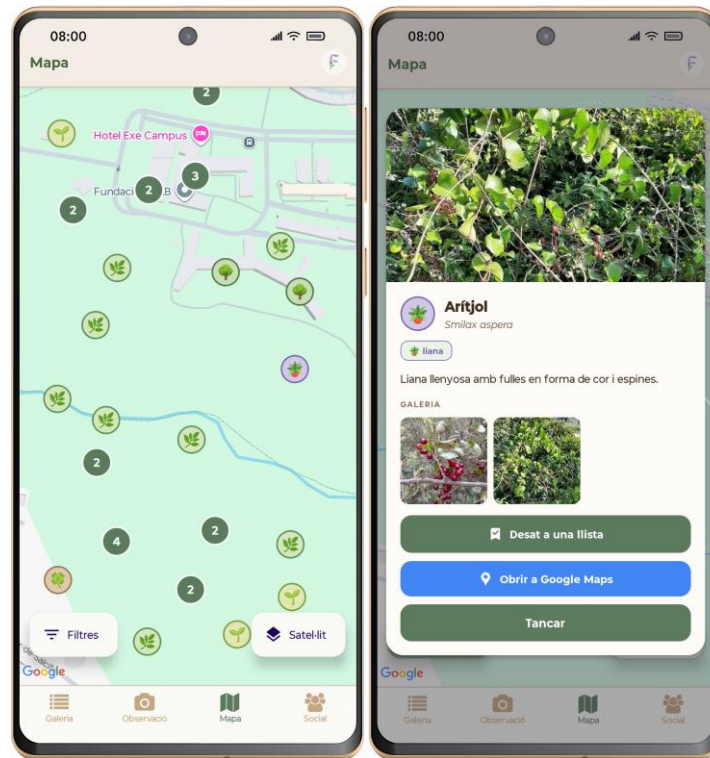


Figure 4: Interactive Geospatial Map showing georeferenced observation markers clustered with custom plant-type icons at the UAB Campus pilot area (left) and the species identification details panel showing description, photo, link to Google Maps (right).

Species Gallery. A high-performance filterable catalog exposes the master species dictionary in a searchable grid format. Filters include growth form (tree, shrub, herb, climber), taxonomic family, protection status, and alien species flag. Each species card presents the primary photograph, scientific name, and common name. The species detail view synthesizes aggregated morphological traits from all validated observations, community-contributed photographs ranked by quality score, and the complete observation history on the campus map.

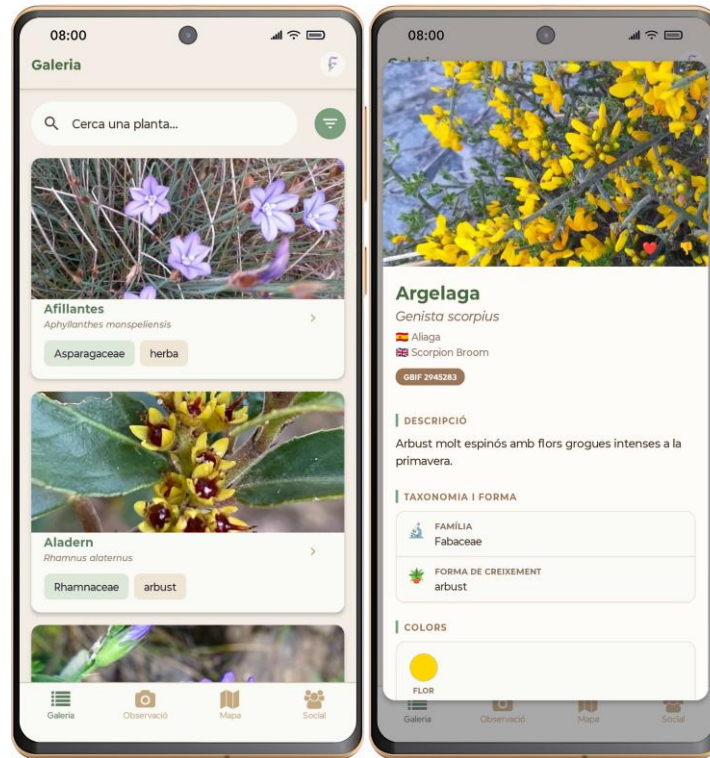


Figure 5: Species Gallery interface, showing the searchable catalog grid with botanical tags and classifications (left) and the detailed species profile sheet displaying multilingual common names, GBIF taxonomy integration, and morphological traits (right).

Observation Capture and Submission Flow. The submission workflow guides the user through five sequential steps:

- High-resolution photograph acquisition via **expo-camera** with haptic feedback confirmation.
- Automatic GPS coordinate capture via **expo-location**.
- AI-assisted species suggestion via PlantNet ⁵, with user review and override capability.
- Morphological trait completion — growth form, leaf characteristics (margin, arrangement), reproductive traits (flower symmetry, plant sex, fruit type), and organ-specific color capture via an integrated eyedropper tool that extracts HEX values from the photograph.
- Submission, which triggers parallel upload of the image to Supabase Storage and insertion of the observation metadata record into PostgreSQL, with offline queuing to SQLite if connectivity is unavailable.

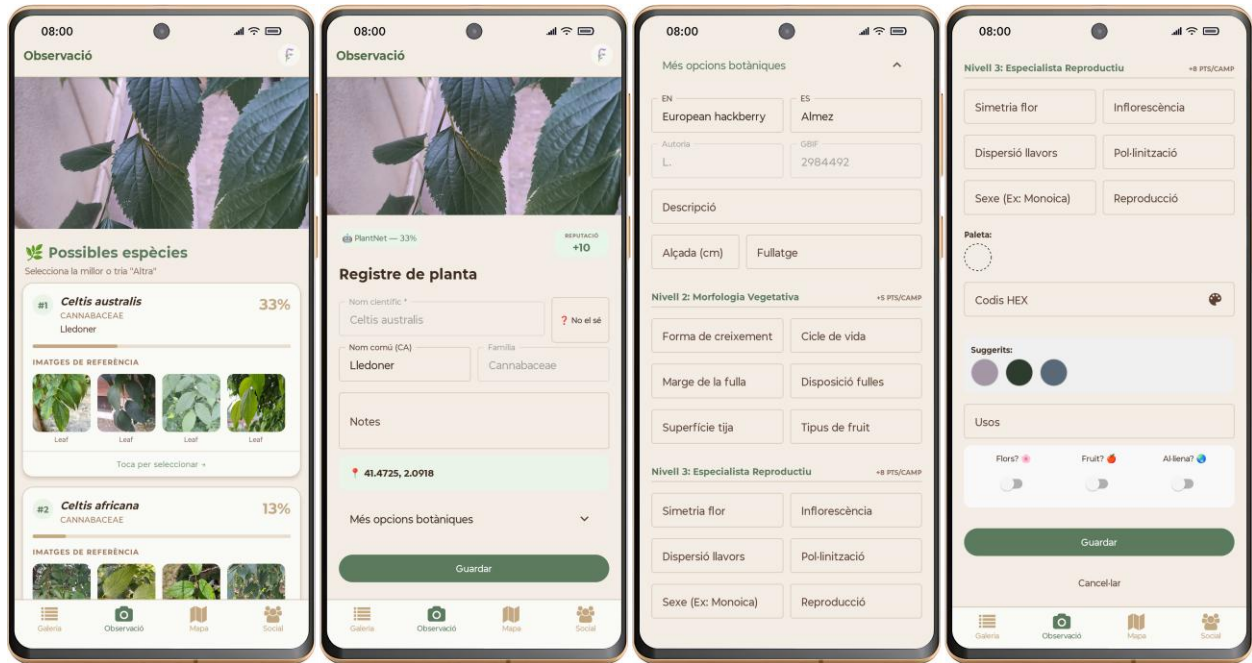


Figure 6: Observation Capture Flow — sequential screens showing camera acquisition, PlantNet AI species suggestion with confidence score, and morphological trait completion form with colour eyedropper.

Authentication and Role Management. Google OAuth and email/password registration are supported. New users are assigned the `student` role (rank_level: 1) on account creation. Role elevation to `expert` or `professor` is performed by an administrator and is immutable by the user. All role-dependent logic — voting weight, moderation impact, observation visibility — is enforced at the database layer via RLS policies, making it tamper-resistant from the client side.

Social Feed and Validation Swiper. The Social tab presents a chronological feed of recent community observations with like/flag controls. The validation swiper presents unvalidated observations for identification review, using a card-swipe UX (affirm right / dispute left) that minimizes friction for rapid community engagement. Each swipe triggers a vote record insertion and recalculates the observation's confidence score in real time.

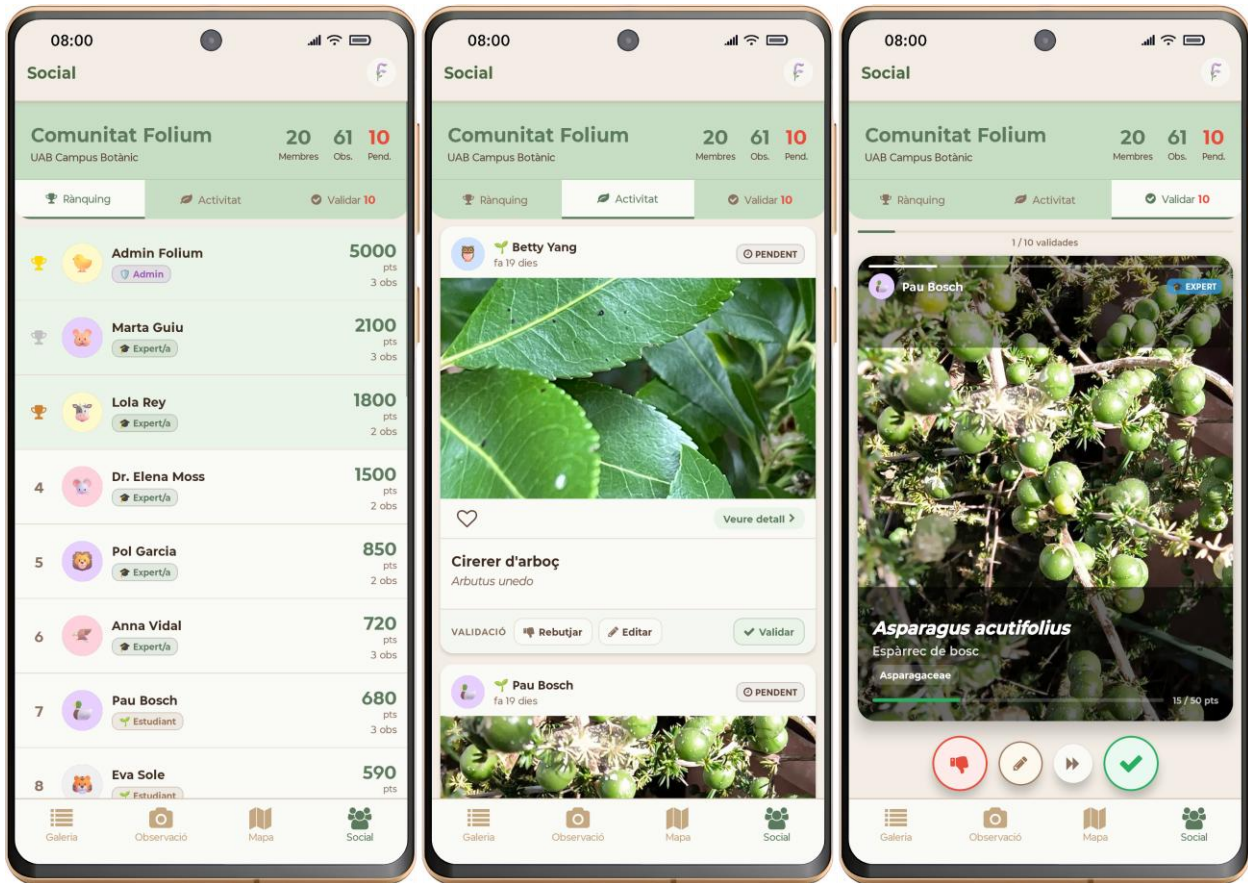


Figure 7: Social Section interface, showing its three main modules: the global community leaderboard with reputation ranking and academic credentials (left), the chronological activity feed of pending observations (center), and the interactive validation swiper (right).

User Profile and Gamification. Each user profile displays cumulative observation count, reputation score, earned achievement badges, personal botanical lists, and rank progression toward the next tier. A global leaderboard ranks community members by reputation score, implementing the competitive incentive structure recommended by the tutor during the first progress review. The system tracks badge triggers (first observation submitted, first validation received, first expert agreement, species diversity milestones) and delivers them via push notification through EAS Notifications.

Account Management and GDPR Tools. Users can edit their public profile, modify credentials, and initiate self-serve account deletion within the Account & Privacy settings. Deletion triggers a Supabase Edge Function that: removes all personally identifying fields from the `profiles` record, deletes the user's avatar from the `user-profiles` bucket, nullifies the `user_id` foreign key in all associated `observations` (preserving scientific records while severing attribution), and invokes the Supabase Auth account deletion endpoint. A confirmation email is dispatched via the Resend transactional email service. This workflow satisfies GDPR Article 17 (Right to Erasure) ⁶ and is a mandatory prerequisite for both Google Play Store and Apple App Store approval.

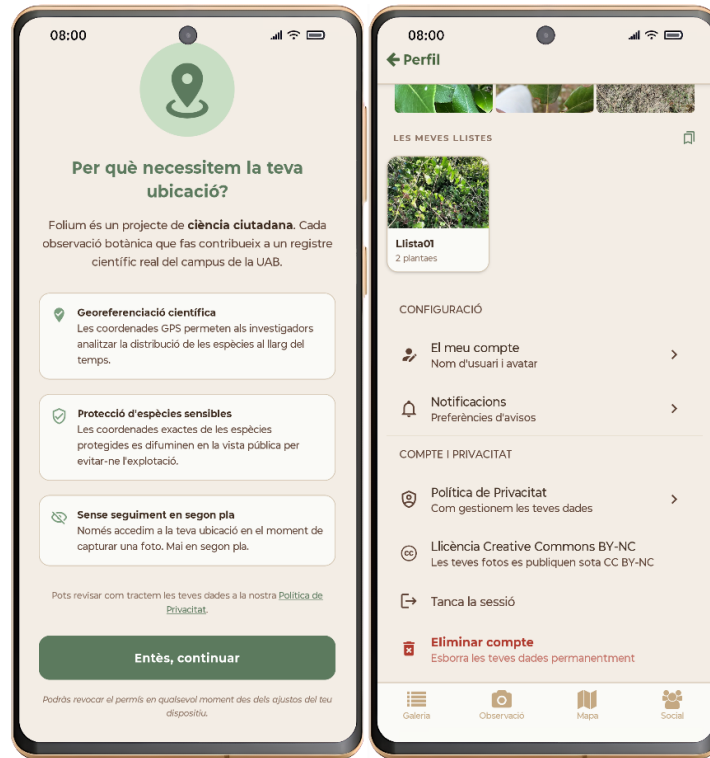


Figure 8: Legal Compliance and Privacy Interfaces, showing the location permission transparency onboarding screen with justification details (left) and the account settings panel featuring privacy policies, Creative Commons license details, and account deletion (right).

4.3. Data Governance: Consensus Validation Model

The epistemic quality of the botanical database is managed through a mathematical governance framework adapted from iNaturalist's Research Grade criteria ¹ and the ICO expert alert model ³. Every validation action contributes a rank-weighted delta to the target observation's `confidence_score`:

- **Student** (rank_level 1): +5 per confirmation, -5 per moderation flag
- **Expert** (rank_level 2): +25 per confirmation, -15 per moderation flag
- **Professor** (rank_level 3): +50 per confirmation (sufficient alone for `scientifically_validated`), -30 per moderation flag

An observation reaches `community_verified` status at `confidence_score ≥ 50` (equivalent to: 10 student votes, 2 expert votes, or 1 professor vote). The `scientifically_validated` status requires either a single professor confirmation or cumulative consensus exceeding the threshold. Observations automatically enter archival status if `moderation_score ≤ -60`, providing spam and misinformation control that scales with community growth without requiring manual administrative review. Individual photographs are independently ranked by `(likes - dislikes)` from `media_votes`; photos falling below -60 are automatically hidden from public display while remaining accessible for administrative audit.

This model provides a continuous spectrum of data confidence rather than a binary valid/invalid classification, enabling downstream users of the botanical dataset to apply their own quality thresholds depending on research requirements.

4.4. Legal and Compliance Framework

GDPR. All user data is stored in Supabase's EU-hosted PostgreSQL instance, satisfying data residency requirements ⁶. The Privacy Policy, publicly hosted at a permanent URL and linked within the application settings, details: categories of personal data collected (name, email, GPS coordinates, photographic content); legal bases for processing; retention periods; user rights (access, erasure, portability, objection); and contact information for data inquiries. Location data is solicited only after a dedicated onboarding screen explains its scientific purpose (geo-attribution of botanical observations), satisfying the contextual permissioning standard required by both the GDPR consent framework and app store review guidelines.

Creative Commons BY-NC 4.0. All user-generated botanical observations — photographs, georeferenced records, morphological trait data — are licensed under Creative Commons Attribution-NonCommercial 4.0 International upon submission ⁷. This license ensures that the accumulated botanical database remains freely accessible for academic and educational use while protecting contributors from unauthorized commercial redistribution. Users retain copyright ownership of their own photographs; the platform receives a non-exclusive, royalty-free, perpetual license to host, display, index, and aggregate the contributed data for the purposes of the citizen science mission.

Content moderation. The `reports` table stores UGC moderation flags submitted by users via an in-app "Report" button available on every observation and photograph. Each flag carries a rank-weighted moderation impact applied to the target's `moderation_score`. This mechanism satisfies the Google Play Store's requirement for a user-accessible inappropriate content reporting system.

4.5. Innovation and Differentiating Factors

Folium advances the state of practice in three substantive dimensions:

Location specificity as a design principle. Existing citizen science platforms treat geographic and community scope as emergent properties of participation. *Folium* treats location specificity as a first-class architectural constraint: the role hierarchy is calibrated to the community's expertise structure, the gamification model rewards the contribution types valued in a place-specific scientific community, and the geographic scope is bounded to a defined polygon. This design philosophy improves data quality (eliminating out-of-scope observations), strengthens community identity (participants share a common place), and enables integration with site-specific spatial layers.

Expert-weighted consensus as an epistemic model. The rank-weighted voting scheme (Student: 5 pts, Expert: 25 pts, Professor: 50 pts) introduces a structured epistemic hierarchy absent from purely democratic platforms. This design allows a credentialed researcher to accelerate the validation of high-confidence observations without eliminating the participatory value of student engagement — a balance critical for sustaining a volunteer contributor base while maintaining scientific data quality.

Production-grade deployment as proof of concept. The project demonstrates that the tooling convergence of managed native frameworks (Expo) ⁸, Backend-as-a-Service (Supabase) ¹⁰, and CI/CD automation (Fastlane + EAS) ¹¹ enables a single undergraduate developer to deliver a production-grade, legally compliant, app-store-distributed citizen

science platform within a one-year academic project at near-zero operational cost. This has significant implications for the feasibility of distributed institutional biodiversity monitoring programs at universities globally.

5. Budget

The project was executed at a total cost of USD 25.00 (Table 2). All recurring infrastructure services operate within their respective free tiers, which are sufficient for pilot-scale deployment. The selective use of free professional tiers rather than self-hosted alternatives eliminates infrastructure management overhead that would be disproportionate for a single-developer academic project, while providing the reliability and scalability guarantees required for a production application.

Table 2: Complete project budget with justification.

Item	Type	Cost	Justification
Google Play Developer registration	One-time	\$25.00	Mandatory for publishing to Google Play Store. Provides permanent developer account; no substitute exists for official store distribution.
Supabase (database, auth, storage, edge functions)	Recurring	\$0.00	Free tier: 500 MB database, 1 GB storage, 50,000 MAU. Sufficient for academic pilot scale. EU data residency satisfies GDPR requirements.
Expo EAS Build & Update	Recurring	\$0.00	Free tier: 30 managed builds/month; unlimited OTA updates. Sufficient for development and pilot phases.
GitHub (version control)	Recurring	\$0.00	Free tier for public and private repositories. Provides Git history, issue tracking, and static documentation hosting.
Resend (transactional email)	Recurring	\$0.00	Free tier: 100 emails/day. Covers account deletion confirmation, validation alert notifications, and system communications.
PlantNet API (species identification)	Per-call	\$0.00	Academic/research use covered under Pl@ntNet open API terms for non-commercial projects.
Apple Developer Program (iOS distribution)	Annual	Not purchased	USD 99.00/year fee excluded from scope; iOS demonstration available via Expo Go Tunnel mode. Formal App Store submission is a post-academic extension.
Total project investment		\$25.00	

The Apple Developer Program was deliberately excluded from the project budget. The Google Play Store provides the primary formal store validation required for the thesis defense. Cross-platform iOS demonstration is achievable without App Store provisioning via Expo Go's Tunnel mode, which allows live device testing from any iOS or Android device during the committee presentation.

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9. Annex A. Supplementary Project Documentation

The following resource is not cited as a primary source in the body of this work but leads to extended technical documentation, implementation details, and additional context about the Folium project that may be useful for committee review or future replication.

A.1. Project Technical Blog

González Bartomeu T. Folium: End-to-End Design and Implementation of a Location-Scoped Citizen Science App for Botanical Cataloging — PoltorProgrammer [Internet]. 2026 [cited 2026 May 20]. Available from:

<https://poltorprogrammer.github.io/post.html#folium>

This post documents the full technical approach behind Folium, covering architectural decisions, implementation trade-offs, and deployment details that fall outside the scope of this academic memory. It serves as a living reference for the project and includes supplementary sources consulted during development.